

Henry Du

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EDUCATION

University of Florida

Doctor of Philosophy in Industrial and Systems Engineering; GPA: 3.75/4.0

Gainesville, FL

Aug. 2023 – Present

University of Delaware

Master of Statistics; GPA: 4.0/4.0; Full Scholarship

Newark, DE

Aug. 2021 – May 2023

Duke University

Master in Environmental Management; GPA: 3.76/4.0

Durham, NC

Aug. 2021 – May 2023

PUBLICATIONS

- [1] **Du, H.**, Liang, M., & Zhong, X. *A Data-Driven Graph Model Approach to Critical Care Patient Clinical Trajectory Prediction*[J]. **IEEE Robotics and Automation Letters**, 2025.
- [2] **Du, H.**, Liang, M., & Zhong, X. *A Unified Framework for Structure-Aware Clustering and Heterogeneous Causal Graph Learning*. **IJSE Transactions**, under revision.

RESEARCH EXPERIENCE

University of Florida

Research Assistant — Penalized DAG-based Clustering Optimization

Gainesville, FL

Sept. 2024 — Sept. 2025

- Developed an end-to-end unsupervised causal-clustering framework using Structural Equation Modeling (SEM) and DAGs to identify latent subpopulations in high-dimensional data.
- Engineered a scalable optimization solver in Python using ADMM and Augmented Lagrangian methods; optimized non-convex objective functions to outperform causal discovery accuracy over baselines.

University of Florida

Research Assistant — Data-Driven Patient Trajectory Prediction

Gainesville, FL

Jan. 2024 — Jan. 2025

- Proposed a two-step pipeline combining soft-DTW clustering and causal discovery to identify ICU patient subgroups and predict clinical trajectories.
- Developed cluster-specific Dynamic SEMs in Python; implemented continuous acyclicity constraints to capture both contemporaneous and time-lagged causal dependencies.

WORK EXPERIENCE

DiDi Global — Anti-Fraud Data Science Intern

Remote — May 2026 – Aug. 2026

- Built an automated driver-audit workflow and coverage dashboard, enabling real-time reviews and increasing full-audit coverage to **over 99.98%**.
- Designed anomalous vehicle-registration monitoring and digital-credential controls, reducing daily active drivers with invalid documents from **600+** to **100+** (over **80%**).
- Engineered a **risk-feature table** and monitoring dashboard for channel-promotion orders, screening **400K+** daily driver-rider interactions, identifying **1,200+** risky drivers and **4,000+** risky passengers, and intercepting **130+** risky activities per day.
- Applied **LLM-assisted feature extraction** to historical fraud samples and developed a **WOE scorecard** with 19 features across 5 dimensions, uncovering **398** high-risk new drivers missed by existing rules.
- Automated daily monitoring reports for background checks, driving reviews, electronic credentials, and risk disposition through **LLM-based workflows**, reducing manual analysis effort.

JPMorgan Chase — Data Scientist Intern, Credit Card Customer Growth

Wilmington, DE — Jun. 2025 – Aug. 2025

- Analyzed conversion in a **100K+** **customer** acquisition pilot to inform segment-level offer strategy.
- Designed and analyzed **A/B tests** of **3 reward offers** across **8 customer segments**, quantifying effects on application rates and incremental applications.
- Built an **XGBoost** response model to forecast application rates and compare predicted with observed campaign performance, supporting campaign evaluation and strategy refinement.

Barclays Bank — Senior Analyst in High Risk Management

Wilmington, DE — Jan. 2023 – Jun. 2023

- Utilized **SAS and SQL** to develop automated monthly waterfall models and risk reports, analyzing key performance drivers across **3 major business lines**.
- Built monthly waterfall models and reports covering **4 business lines** assessing key risk drivers.
- Developed policy simulations using **SAS and SQL**, quantifying the marginal impact of tightening closure thresholds.

Barclays Bank — Senior Analyst in Credit Card Product

Wilmington, DE — Jun. 2022 – Jan. 2023

- Built an **Oracle-based** portfolio monitoring framework tracking **20+ KPIs** with trend views to support teams.
- Developed and maintained **3 interactive Tableau dashboards** with Oracle, reducing processing time by **30%**.
- Designed **Tableau** dashboards by co-brand portfolio to analyze acquisition-channel effectiveness, supporting budget allocation and winning 1st Prize in the company.

SKILLS

Technical Skills: Python, R, Java, SQL, HiveSQL, SAS, Spark, PHP, HTML, Tableau, JMP, Git, Oracle, Gurobi